

A cross-alphabet identity for bounded-difference square arrays: OEIS A125539

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Abstract

OEIS A125539 carries an empirical identity relating its counts to those of the same problem over an alphabet one smaller. It is true, and sharply so: the threshold the entry states is exactly the diameter bound of the grid graph, and the identity fails below it. The proof is four lines and uses no machinery.

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1 The conjecture

OEIS A125539 is “Number of $n \times n$ arrays with entries in $1..8$ in which adjacent entries differ by 5 or less (adjacent means in x or y directions).”. It has offset 0 and begins

1, 8, 2902, 52127130, 45120225704486, 1888405717902201691046, 3819825084723229625771326323002, 373453238

The entry states, as a conjecture contributed by R. H. Hardin and never marked settled:

[Empirical] $a(\text{base}+1, n, \text{diff}) = a(\text{base}, n, \text{diff}) + F(n, \text{diff})$ for $\text{base} \geq 2 \cdot \text{diff} \cdot (n-1)$ and $F(n, \text{diff}) =$
($n=2$, A063496($\text{diff}+1$)) ($n=3$, A068744($\text{diff}+1$)) ($n=4$, A068745($\text{diff}+1$)) ($n=5$, A068746($\text{diff}+1$))
($n=6$, A068747($\text{diff}+1$))

As of the “Last modified” line on the live entry (Mar 31 2012 12:35:05, revision 6) this is still recorded as empirical. For this entry’s own alphabet the threshold $2d(n-1) \leq b$ leaves only $n=1$, so the identity is not a statement about this entry’s published terms beyond the first; it is a statement about the family, and it is the family statement that is proved below.

2 Notation

For integers $b \geq 1$, $d \geq 1$ and $n \geq 1$ write $T(b, d, n)$ for the number of $n \times n$ arrays with entries in $\{1, \dots, b\}$ in which any two orthogonally adjacent entries differ by at most d . The entry is $T(8, 5, n)$, and the conjecture is

$$T(b+1, d, n) = T(b, d, n) + F(n, d) \quad \text{for } b \geq 2d(n-1),$$

with $F(n, d)$ the quantity the entry tabulates through its reference sequences ($n=2$: OEIS A063496, $n=3$: OEIS A068744, $n=4$: OEIS A068745, $n=5$: OEIS A068746, $n=6$: OEIS A068747).

3 The proof

Theorem 1. For all $n \geq 1$, $d \geq 1$ and $b \geq 2d(n-1)$,

$$T(b+1, d, n) - T(b, d, n) = F(n, d),$$

where $F(n, d)$ is the number of admissible arrays over $\mathbb{Z}$ whose maximum entry is 0 — a quantity depending on n and d only.

Proof. Step 1. An admissible array over $\{1, \dots, b+1\}$ either uses the value $b+1$ or does not, and those that do not are exactly the admissible arrays over $\{1, \dots, b\}$. So $T(b+1, d, n) - T(b, d, n)$ counts the admissible arrays over $\{1, \dots, b+1\}$ that use $b+1$.

Step 2. Every admissible array satisfies $\max - \min \leq 2d(n-1)$. Let u attain the maximum and v the minimum. In the grid graph on $n \times n$ cells with orthogonal adjacency any two cells are joined by a path of length at most $2(n-1)$ — move along a row, then along a column — and along a path of length L the entry changes by at most dL , since each step is between adjacent cells.

Step 3. An array counted in Step 1 has maximum exactly $b+1$, so by Step 2 all of its entries lie in $[b+1-2d(n-1), b+1]$. When $b \geq 2d(n-1)$ that interval is contained in $\{1, \dots, b+1\}$, so the constraint “entries in $\{1, \dots, b+1\}$ ” is implied by the maximum and the range bound and imposes nothing further. The arrays counted in Step 1 are therefore in bijection — by the translation carrying the maximum to 0 — with the admissible arrays over $\mathbb{Z}$ of maximum 0, and that count involves no b . $\square$

Remark 2. The threshold is sharp, and not merely sufficient. Below it the window no longer fits and the identity fails: at $b = 5$, $d = 3$, $n = 3$ the difference $T(6, 3, 3) - T(5, 3, 3)$ is 964,755 while $F(3, 3) = 1,253,329$.

4 Verification

Four checks, each independent of the algebra above.

First, the count $T(8, 5, n)$ was computed by an independent row-pair dynamic program and reproduces the 4 published terms of A125539 that the computation reaches, exactly, in integer arithmetic.

Second, the identity was evaluated directly at the entry’s own alphabet size for every n the threshold covers, and at one n beyond it, where it fails as the remark says.

Third, $F(n, d)$ as computed here agrees with the entry’s own reference sequences. The entry writes them as “A...(*diff* + 1)”, and the index is positional — the (*diff* + 1)st term as listed. That reading is forced: the sequences do not share an offset, and only the positional reading makes $n = 2$ and $n = 3$ agree simultaneously.

Fourth, the following table was produced by the same program and can be checked by hand at small n :

n	$T(8+1, 5, n)$	$T(8, 5, n)$	$F(n, 5)$
1	9	8	1
2	3773	2902	891

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A125539: <https://oeis.org/A125539>.